

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5423A

5 Credits: 4

Program Elective

Source-grounded

Food chemistry II

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5423A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5423A.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Food Processing Technology
Course code	5423A
Course title	Food chemistry II
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4(L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

12 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	17 Understanding modification techniques such as hydrogenation and inter-esterification, and the use of fat replacements. Describe the types of food allergies, the chemistry of	See official syllabus
CO2	food allergens, mechanisms of allergic reactions, and 13 Understanding allergenic components in foods. Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination	See official syllabus
CO3	of toxicants, and assessment of food safety, including 17 Understanding risk assessment, acute toxicity, and various toxicological effects. Describe toxic substances in food and their toxic	See official syllabus
CO4	13 Understanding effects in people	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	inter-esterification, and the use of fat replacements.	3	As specified
Module 2	Describe the types of food allergies, the chemistry of food allergens,	3	As specified
Module 3	safety, including risk assessment,acute toxicity, and various toxicological	3	As specified
Module 4	Describe toxic substances in food and their toxic effects in people	3	As specified

MODULE 1

inter-esterification, and the use of fat replacements.

This module is presented from the official syllabus scope for Food chemistry II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: inter-esterification, and the use of fat replacements.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Explain dextrins and dextrans, Starch 6 Understanding hydrolysis,

Explain dextrins and dextrans, Starch 6 Understanding hydrolysis, is an official topic in Module 1 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain dextrins and dextrans, Starch 6 Understanding hydrolysis, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain dextrins and dextrans, starch 6 understanding hydrolysis, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain dextrins and dextrans, Starch 6 Understanding hydrolysis, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain dextrins and dextrans, Starch 6 Understanding hydrolysis, definition/meaning. steps, application. Technical terms English-

▼ Discuss Modification of fats 6 Understanding Discuss fat replacements

Discuss Modification of fats 6 Understanding Discuss fat replacements is an official topic in Module 1 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss Modification of fats 6 Understanding Discuss fat replacements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss modification of fats 6 understanding discuss fat replacements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Discuss Modification of fats 6 Understanding Discuss fat replacements means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Discuss Modification of fats 6 Understanding Discuss fat replacements definition/meaning . . . , steps, application Technical terms English-

▼ 5 Understanding cis and trans isomers, inter-esterification, Fat replacements.

5 Understanding cis and trans isomers, inter-esterification, Fat replacements. is an official topic in Module 1 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding cis and trans isomers, inter-esterification, Fat replacements. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding cis and trans isomers, inter-esterification, fat replacements. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 5 Understanding cis and trans isomers, inter-esterification, Fat replacements. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 5 Understanding cis and trans isomers, inter-esterification, Fat replacements. definition/meaning, steps, application. Technical terms English- Malayalam.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Describe the types of food allergies, the chemistry of food allergens,

This module is presented from the official syllabus scope for Food chemistry II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the types of food allergies, the chemistry of food allergens,
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Discuss the chemistry and types of food allergens. 4 Understanding Explain the Mechanisms of allergic-type reactions Understanding

Discuss the chemistry and types of food allergens. 4 Understanding
Explain the Mechanisms of allergic-type reactions Understanding is an official topic in Module 2 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the chemistry and types of food allergens. 4 Understanding Explain the Mechanisms of allergic-type reactions Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the chemistry and types of food allergens. 4 understanding explain the mechanisms of allergic-type reactions understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the chemistry and types of food allergens. 4 Understanding Explain the Mechanisms of allergic-type reactions Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Discuss the chemistry and types of food allergens. 4 Understanding Explain the Mechanisms of allergic-type reactions Understanding definition/meaning steps, application Technical terms English-

▼ 4 Illustrate food allergy and hypersensitivity, Understanding

4 Illustrate food allergy and hypersensitivity, Understanding is an official topic in Module 2 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Illustrate food allergy and hypersensitivity, Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 illustrate food allergy and hypersensitivity, understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Illustrate food allergy and hypersensitivity, Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 4 Illustrate food allergy and hypersensitivity, Understanding definition/meaning. steps, application. Technical terms English- .

▼ 4 mechanisms of allergic-type reactions and allergenic components in foods Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination of toxicants, and assessment of food

4 mechanisms of allergic-type reactions and allergenic components in foods Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination of toxicants, and assessment of food is an official topic in Module 2 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 mechanisms of allergic-type reactions and allergenic components in foods Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination of toxicants, and assessment of food using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 mechanisms of allergic-type reactions and allergenic components in foods explain the methods for quantitative and qualitative

analysis of toxicants in foods, biological determination of toxicants, and assessment of food should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 mechanisms of allergic-type reactions and allergenic components in foods Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination of toxicants, and assessment of food means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 4 mechanisms of allergic-type reactions and allergenic components in foods Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination of toxicants, and assessment of food
 definition/meaning
 . steps, application
 . Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

safety, including risk assessment, acute toxicity, and various toxicological

This module is presented from the official syllabus scope for Food chemistry II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: safety, including risk assessment, acute toxicity, and various toxicological
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ toxicants in foods

toxicants in foods is an official topic in Module 3 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify toxicants in foods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, toxicants in foods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What toxicants in foods means in the official course context
Study lens	Principle → components or stages → result → application

[illegible]

▼ Explain biological determination of toxicants 6 Understanding Summarize assessment of food

Explain biological determination of toxicants 6 Understanding Summarize assessment of food is an official topic in Module 3 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain biological determination of toxicants 6 Understanding Summarize assessment of food using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain biological determination of toxicants 6 understanding summarize assessment of food should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain biological determination of toxicants 6 Understanding Summarize assessment of food means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain biological determination of toxicants 6 Understanding Summarize assessment of food
 definition/meaning .
 , steps, application . Technical terms English-
 .

▼ 3 safety Understanding determination of toxicants, Assessment of food safety – Risk assessment and risk benefit indices of human exposure, acute toxicity, mutagenicity and carcinogenicity, reproductive and developmental toxicity, neurotoxicity and behavioral effect, immunotoxicity

3 safety Understanding determination of toxicants, Assessment of food safety – Risk assessment and risk benefit indices of human exposure, acute toxicity, mutagenicity and carcinogenicity, reproductive and developmental toxicity, neurotoxicity and behavioral effect, immunotoxicity is an official topic in Module 3 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 safety Understanding determination of toxicants, Assessment of food safety – Risk assessment and risk benefit indices of human exposure, acute toxicity, mutagenicity and carcinogenicity, reproductive and developmental toxicity, neurotoxicity and behavioral effect, immunotoxicity using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 safety understanding determination of toxicants, assessment of food safety – risk assessment and risk benefit indices of human exposure, acute toxicity, mutagenicity and carcinogenicity, reproductive and developmental toxicity, neurotoxicity and behavioral effect, immunotoxicity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 safety Understanding determination of toxicants, Assessment of food safety – Risk assessment and risk benefit indices of human exposure, acute toxicity, mutagenicity and carcinogenicity, reproductive and developmental toxicity, neurotoxicity and behavioral effect, immunotoxicity means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 safety Understanding determination of toxicants, Assessment of food safety – Risk assessment and risk benefit indices of human exposure, acute toxicity, mutagenicity and carcinogenicity, reproductive and developmental toxicity, neurotoxicity and behavioral effect, immunotoxicity എന്നിങ്ങനെയാണ് പദം definition/meaning എന്നിങ്ങനെ. പദം പദം പദം, steps, application എന്നിങ്ങനെ എന്നിങ്ങനെ. Technical terms English- പദം പദം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Describe toxic substances in food and their toxic effects in people

This module is presented from the official syllabus scope for Food chemistry II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe toxic substances in food and their toxic effects in people
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Discuss Food additives as toxicants in food

Discuss Food additives as toxicants in food is an official topic in Module 4 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss Food additives as toxicants in food using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss food additives as toxicants in food should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss Food additives as toxicants in food means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Discuss Food additives as toxicants in food definition/meaning . steps, application . Technical terms English- .

▼ Explain toxicants formed during food processing 4 Understanding Outline the risk of genetically modified food,

Explain toxicants formed during food processing 4 Understanding Outline the risk of genetically modified food, is an official topic in Module 4 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain toxicants formed during food processing 4 Understanding Outline the risk of genetically modified food, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain toxicants formed during food processing 4 understanding outline the risk of genetically modified food, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain toxicants formed during food processing 4 Understanding Outline the risk of genetically modified food, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain toxicants formed during food processing 4 Understanding Outline the risk of genetically modified food, definition/meaning . steps, application . Technical terms English- .

▼ food supplements, persistent organic pollutants 4 Understanding formed during food processing such as nitrosamines, maillard reaction products acrylamide, benzene, and irradiation; risk of genetically modified food, food supplements, persistent organic pollutants, Text / Reference T/R Book Title/Author R1 Concon JM.1988. Food Toxicology - Principles & Concepts. Marcel Dekker. R2 Steven T. 1989. Food Toxicology: A Perspective on Relative Risks R3 Helferich, William and Carl K.Winter "Food Toxicology" CRC Press, 2001 R4 Shibamoto, Taka yuki and Leonard F.Bjeldanzes "Introduction to Food Toxicology" 2nd Edition.Academic Press, 2009. R5 Shibamoto T. and Bjeldanes L., Introduction to Food Toxicology, Academic Press, Inc. San Diego, CA. ISBN 0-12640025-3 2. Deshpande, S.S. (2002). R6 Handbook of Food Toxicology, Marcel Dekker Inc. NY ISBN 0- 8247-0760-5 R7 Stine, K. and Brown, T. 1996. Principles of Toxicology. CRC Press, Inc. Boca Raton, FL. R8 Püssa, Tõnu. 2007. Principles of Food Toxicology. CRC Press, LLC. Boca Raton, FL. 5. Online Resources Sl.No Website Link 1 <https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst-128-food-toxicology> 2 <https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens> 3 <https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food> 4 <https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food> 5 <https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600>

food supplements, persistent organic pollutants 4 Understanding formed during food processing such as nitrosamines, maillard reaction products acrylamide, benzene, and irradiation; risk of genetically modified food, food supplements, persistent organic pollutants, Text / Reference T/R Book Title/Author R1 Concon JM.1988. Food Toxicology - Principles & Concepts. Marcel Dekker. R2 Steven T. 1989. Food Toxicology: A Perspective on Relative Risks R3 Helferich, William and Carl K.Winter "Food Toxicology" CRC Press, 2001 R4 Shibamoto, Taka yuki and Leonard F.Bjeldanzes "Introduction to Food Toxicology" 2nd Edition.Academic Press, 2009. R5

Shibamoto T. and Bjeldanes L., Introduction to Food Toxicology, Academic Press, Inc. San Diego, CA. ISBN 0-12640025-3

2. Deshpande, S.S. (2002). R6 Handbook of Food Toxicology, Marcel Dekker Inc. NY ISBN 0- 8247-0760-5

R7 Stine, K. and Brown, T. 1996. Principles of Toxicology. CRC Press, Inc. Boca Raton, FL.

R8 Püssa, Tõnu. 2007. Principles of Food Toxicology. CRC Press, LLC. Boca Raton, FL.

5. Online Resources

Sl.No Website Link

1 <https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst-128-food-toxicology>

2 <https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens>

3 <https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food>

4 <https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food>

5 <https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600>

is an official topic in Module 4 of Food chemistry II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify food supplements, persistent organic pollutants
 - 4 Understanding formed during food processing such as nitrosamines, maillard reaction products acrylamide, benzene, and irradiation; risk of genetically modified food, food supplements, persistent organic pollutants,
- Text / Reference T/R Book Title/Author
- R1 Concon JM.1988. Food Toxicology - Principles & Concepts. Marcel Dekker.
- R2 Steven T. 1989. Food Toxicology: A Perspective on Relative Risks
- R3 Helferich, William and Carl K.Winter "Food Toxicology" CRC Press, 2001
- R4 Shibamoto, Taka yuki and Leonard F.Bjeldanes "Introduction to Food Toxicology" 2nd Edition.Academic Press, 2009.
- R5 Shibamoto T. and Bjeldanes L., Introduction to Food Toxicology,

Academic Press, Inc. San Diego, CA. ISBN 0-12640025-3 2. Deshpande, S.S. (2002). R6 Handbook of Food Toxicology, Marcel Dekker Inc. NY ISBN 0-8247-0760-5 R7 Stine, K. and Brown, T. 1996. Principles of Toxicology. CRC Press, Inc. Boca Raton, FL. R8 Püssa, Tõnu. 2007. Principles of Food Toxicology. CRC Press, LLC. Boca Raton, FL. 5. Online Resources Sl.No Website Link 1 <https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst-128-food-toxicology> 2 <https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens> 3 <https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food> 4 <https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food> 5 <https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, food supplements, persistent organic pollutants 4 understanding formed during food processing such as nitrosamines, maillard reaction products acrylamide, benzene, and irradiation; risk of genetically modified food, food supplements, persistent organic pollutants, text / reference t/r book title/author r1 concon jm.1988. food toxicology - principles & concepts. marcel dekker. r2 steven t. 1989. food toxicology: a perspective on relative risks r3 helferich, william and carl k.winter "food toxicology" crc press, 2001 r4 shibamoto, taka yuki and leonard f.bjeldanzes "introduction to food toxicology" 2nd edition.academic press, 2009. r5 shibamoto t. and bjeldanes l., introduction to food toxicology, academic press, inc. san diego, ca. isbn 0-12640025-3 2. deshpane, s.s. (2002). r6 handbook of food toxicology, marcel dekker inc. ny isbn 0- 8247-0760-5 r7 stine, k. and brown, t. 1996. principles of toxicology. crc press, inc. boca raton, fl. r8 püssa, tõnu. 2007. principles of food toxicology. crc press, llc. boca raton, fl. 5. online resources sl.no website link 1 <https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst->

128-food-toxicology 2 <https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens> 3 <https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food> 4 <https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food> 5

<https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What food supplements, persistent organic pollutants 4 Understanding formed during food processing such as nitrosamines, maillard reaction products acrylamide, benzene, and irradiation; risk of genetically modified food, food supplements, persistent organic pollutants, Text / Reference T/R Book Title/Author R1 Concon JM.1988. Food Toxicology - Principles & Concepts. Marcel Dekker. R2 Steven T. 1989. Food Toxicology: A Perspective on Relative Risks R3 Helferich, William and Carl K.Winter "Food Toxicology" CRC Press, 2001 R4 Shibamoto, Taka yuki and Leonard F.Bjeldanzes "Introduction to Food Toxicology" 2nd Edition.Academic Press, 2009. R5 Shibamoto T. and Bjeldanes L., Introduction to Food Toxicology, Academic Press, Inc. San Diego, CA. ISBN 0-12640025-3 2. Deshpande, S.S. (2002). R6 Handbook of Food Toxicology, Marcel Dekker Inc. NY ISBN 0- 8247-0760-5 R7 Stine, K. and Brown, T. 1996. Principles of Toxicology. CRC Press, Inc. Boca Raton, FL. R8 Püssa, Tõnu. 2007. Principles of Food Toxicology. CRC Press, LLC. Boca Raton, FL. 5. Online Resources SI.No Website Link 1 https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst-128-food-toxicology 2 https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens 3 https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food 4 https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food 5 https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- food supplements, persistent organic pollutants 4 Understanding formed during food processing such as nitrosamines, maillard reaction products acrylamide, benzene, and irradiation; risk of genetically modified food, food supplements, persistent organic pollutants, Text / Reference T/R Book Title/Author R1 Concon JM.1988. Food Toxicology - Principles & Concepts. Marcel Dekker. R2 Steven T. 1989. Food Toxicology: A Perspective on Relative Risks R3 Helferich, William and Carl K.Winter "Food Toxicology" CRC Press, 2001 R4 Shibamoto, Taka yuki and Leonard F.Bjeldanzes "Introduction to Food Toxicology" 2nd Edition.Academic Press, 2009. R5 Shibamoto T. and Bjeldanes L., Introduction to Food Toxicology, Academic Press, Inc. San Diego, CA. ISBN 0-12640025-3 2. Deshpande, S.S. (2002). R6 Handbook of Food Toxicology, Marcel Dekker Inc. NY ISBN 0- 8247-0760-5 R7 Stine, K. and Brown, T. 1996. Principles of Toxicology. CRC Press, Inc. Boca Raton, FL. R8 Püssa, Tõnu. 2007. Principles of Food Toxicology. CRC Press, LLC. Boca Raton, FL. 5. Online Resources Sl.No Website Link 1 <https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst-128-food-toxicology> 2 <https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens> 3 <https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food> 4 <https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food> 5 <https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600> ഫുഡ് ടോക്സിക്കോളജി ഫുഡ് ടോക്സിക് definition/meaning ഫുഡ് ടോക്സിക് ഫുഡ് ടോക്സിക്, steps, application ഫുഡ് ടോക്സിക് ഫുഡ് ടോക്സിക്. Technical terms English-ഫുഡ് ടോക്സിക് ഫുഡ് ടോക്സിക്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

of food safety, including 17 Understanding risk assessment, acute toxicity, and various toxicological effects.						
Describe toxic substances in food and their toxic						
C04	13					
Understanding effects in people						
C0–P0 Mapping						
Course						
	P01	P02	P03	P04	P05	P06
P07						
Outcomes						
C01	2					
C02	2					
C03	2					
C04	2					
3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped						
Course Outline:						
Module Duration						
Cognitive						
Description						
Outcomes (Hours)						

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain Explain dextrins and dextrans, Starch 6 Understanding hydrolysis,. (3 marks)

► Define or explain Discuss Modification of fats 6 Understanding Discuss fat replacements. (3 marks)

► Define or explain 5 Understanding cis and trans isomers, inter-esterification, Fat replacements.. (3 marks)

Module 2

► Define or explain Discuss the chemistry and types of food allergens. 4 Understanding Explain the Mechanisms of allergic-type reactions Understanding. (3 marks)

► Define or explain 4 Illustrate food allergy and hypersensitivity, Understanding. (3 marks)

► Define or explain 4 mechanisms of allergic-type reactions and allergenic components in foods Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination of toxicants, and assessment of food. (3 marks)

Module 3

► **Define or explain toxicants in foods. (3 marks)**

► **Define or explain Explain biological determination of toxicants 6
Understanding Summarize assessment of food. (3 marks)**

► **Define or explain 3 safety Understanding determination of toxicants,
Assessment of food safety - Risk assessment and risk benefit indices of
human exposure, acute toxicity, mutagenicity and carcinogenicity,
reproductive and developmental toxicity, neurotoxicity and behavioral
effect, immunotoxicity. (3 marks)**

Module 4

► **Define or explain Discuss Food additives as toxicants in food. (3
marks)**

► **Define or explain Explain toxicants formed during food processing 4
Understanding Outline the risk of genetically modified food,. (3 marks)**

► **Define or explain food supplements, persistent organic pollutants 4
Understanding formed during food processing such as nitrosamines,
maillard reaction products acrylamide, benzene, and irradiation; risk of
genetically modified food, food supplements, persistent organic
pollutants, Text / Reference T/R Book Title/Author R1 Concon JM.1988.
Food Toxicology - Principles & Concepts. Marcel Dekker. R2 Steven T.
1989. Food Toxicology: A Perspective on Relative Risks R3 Helferich,
William and Carl K.Winter "Food Toxicology" CRC Press, 2001 R4
Shibamoto, Taka yuki and Leonard F.Bjeldanzes "Introduction to Food
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K. and Brown, T. 1996. Principles of Toxicology. CRC Press, Inc. Boca
Raton, FL. R8 Püssa, Tõnu. 2007. Principles of Food Toxicology. CRC**

Press, LLC. Boca Raton, FL. 5. Online Resources Sl.No Website Link 1

<https://foodscience.ucdavis.edu/academic->

[programs/undergraduate/courses/fst- 128-food-toxicology 2](https://foodscience.ucdavis.edu/academic-)

<https://www.foodallergy.org/living-food-allergies/food-allergy->

[essentials/common-allergens 3 https://www.fda.gov/food/chemical-](https://www.foodallergy.org/living-food-allergies/food-allergy-)

[contaminants-pesticides/natural-toxins-food 4](https://www.fda.gov/food/chemical-)

<https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food>

[5 https://www.futurelearn.com/info/courses/nutrition-for-](https://www.futurelearn.com/info/courses/nutrition-for-)

[health/0/steps/213600. \(3 marks\)](https://www.futurelearn.com/info/courses/nutrition-for-)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain dextrins and dextrans, Starch 6 Understanding hydrolysis,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Discuss the chemistry and types of food allergens. 4 Understanding Explain the Mechanisms of allergic-type reactions Understanding.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **toxicants in foods.**

3-mark explanation: Explain one principle, process or comparison from

Module 4

1-mark definition: State the meaning of **Discuss Food additives as toxicants in food.**

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES**References and web resources****References listed in the official PDF**

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website
- <https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst-food-toxicology>
- <https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens>
- <https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food>
- <https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food>
- <https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5423A. Verify institutional updates against the current official curriculum.